1. Clinical Question & Core Hypothesis

Clinical Question

Generated (synthetic) TRUS images can distinguish benign vs malignant lesions, while PSA density and PSA dynamics stratify prostate disease risk — yet a high-risk molecular subset (tumors with truncating NFE2 alterations) shows poor immunotherapy responsiveness. Why might image-based triage fail to identify molecularly immune-resistant, PSA-dense tumors? 2) Tier 1 paths used: (a) PSA density as a robust predictor of disease reclassification and PSA-linked recurrence risk, and (b) NFE2 truncating mutation/upregulation marking a high-risk, immunotherapy-resistant cohort. These provide a direct mechanistic axis linking imaging-derived classification, PSA biology, and a molecular driver of immune resistance. 3) Tier 2 evidence supporting the claim: MSI/dMMR status predicts immunotherapy targetability (so molecular testing can rescue misclassification), PSA levels and biochemical-recurrence link PSA dynamics to outcomes, and MetS/bladder-cancer context shows comorbidity/domain effects that could confound image–biomarker relationships. 4) Endpoint choice rationale: I selected therapy_response (immunotherapy responsiveness proxy) because the path explicitly connects a molecular driver (NFE2) and a high-risk cohort to reduced immunotherapy responsiveness — this yields a clinically actionable question about treatment selection. 5) What I excluded and why: I excluded descriptive protocol/section details (Section 5.2.2, human-observer realism methods), Barrett's esophagus/polyp specifics, and broader chemotherapy exposures to keep the hypothesis sharply focused on the interplay between synthetic imaging, PSA-based stratification, and NFE2-driven immunotherapy resistance?

Working Hypothesis

Primary hypothesis: Synthetic TRUS image realism does not faithfully capture the microarchitectural stromal and vascular features of prostate tumors with truncating NFE2 alterations, particularly in cases with elevated PSA density, leading to under-recognition of an immunosuppressive tumor microenvironment and poorer therapy_response to immunotherapy. Corollary hypotheses: (1) Incorporating PSA density and NFE2 mutation status alongside synthetic TRUS realism will improve prediction of immunotherapy response beyond imaging alone; (2) The predictive gain of the integrated approach is moderated by MSI/dMMR status and metabolic/clinical context (e.g., MetS, PSA levels). Mechanistic rationale: NFE2 truncation can alter redox signaling and immune infiltration, PSA density reflects tumor burden and microenvironmental changes, and synthetic imaging may miss microvascular/stromal cues that correlate with immune resistance. This yields a testable framework to assess whether a combined imaging-genomic approach better identifies patients unlikely to benefit from immune checkpoint therapy.

Element	Value
Imaging feature	Synthetic TRUS images (generated images)
Anchor gene(s)	NFE2, PSA density
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

NFE2: NFE2) and a high-risk cohort to reduced immunotherapy responsiveness — this yields a clinically actionable question about treatment selection.

PSA density: PSA density as a robust predictor of disease reclassification and PSA-linked recurrence risk, and (b) NFE2 truncating mutation/upregulation marking a high-risk, immunotherapy-resistant cohort.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MSI/dMMR status	Supporting modifier from knowledge graph
Secondary	PSA levels	Supporting modifier from knowledge graph
Secondary	biochemical recurrence	Supporting modifier from knowledge graph
Secondary	MetS	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MSI/dMMR status: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PSA levels: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.